

Perimeters and areas of similar figures



- 1 a 18 cm^2 b 72 cm^2 c $1 : 4$ d Yes
- 2 a $\frac{3}{7}$ b $\frac{9}{49}$
- 3 12
- 4 a 36 b 504 cm^2
- 5 a $81 : 169$ b 169 cm^2
- 6 a 3
b 21 cm, 30 cm, 48 cm
c The perimeter of the larger triangle is 3 times larger than the perimeter of the smaller triangle.
- 7 324 cm^2
- 8 $49 : 4$
- 9 It is 9 times the original area.
- 10 a 128 cm^2 b 9 cm
- 11 600 cm^2
- 12 $\frac{25}{16}$
- 13 a 4 b 16 c 24

14

a A will increase by a factor of 4 in total.



b A will remain unchanged.

15

a A will increase by a factor of 9.

b A will decrease by a factor of 16.

Additional questions

16 $\frac{36}{25}$ times bigger

17 6 m^2

18 The enlarged pentagon has a perimeter that is 3.25 times greater than the original pentagon.